

DZHARDZHAN, Russian Federation

WMO: 241430

Lat: **68.735N**Lon: **124.021E**Elev: **39**StdP: **100.86**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.7	-48.0	-51.8	0.0	-49.3	-50.1	0.0	-47.8	17.1	-26.3	14.1	-25.6	0.6	20	0.858

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	12.7	21.9	16.3	11.6	20.7	14.6	10.4	19.0	55.9	24.3	51.4	22.2	46.8	20.7	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.7	11.5	8.9	DB	-52.4	28.9	1.6	1.8	-53.6	30.2	-54.5	31.3	-55.4	32.3	-56.6	33.6	(4)
(5)			WB	-52.4	21.6	1.6	1.5	-53.5	22.7	-54.4	23.5	-55.3	24.4	-56.4	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.1	-36.6	-34.8	-22.4	-8.7	1.0	11.8	14.9	11.9	3.7	-10.3	-28.6	-36.6
	DBStd	20.16	7.07	7.36	8.36	6.55	5.90	4.92	4.32	3.44	3.79	7.20	7.69	7.23
	HDD10.0	8028	1443	1254	1005	561	284	36	5	18	191	629	1158	1443
	HDD18.3	10764	1702	1488	1263	811	538	202	124	201	438	888	1408	1702
	CDD10.0	333	0	0	0	0	4	91	158	77	3	0	0	0
	CDD18.3	27	0	0	0	0	0	6	19	2	0	0	0	0
	CDH23.3	216	0	0	0	0	2	52	148	14	0	0	0	0
	CDH26.7	38	0	0	0	0	0	8	29	0	0	0	0	0
Wind	WSAvg	3.0	2.9	2.3	2.6	3.3	3.5	3.4	3.4	3.1	3.3	3.5	2.4	2.5
Precipitation	PrecAvg	306	13	11	13	13	21	40	50	46	39	38	19	15
	PrecMax	429	43	30	36	42	60	91	136	135	86	82	50	40
	PrecMin	187	3	2	2	0	2	3	9	2	8	6	4	2
	PrecStd	70	8	7	9	9	14	23	31	30	18	17	10	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-16.6	-13.5	-0.8	8.5	19.7	28.0	29.6	25.2	18.1	2.0	-6.9	-16.9
		MCWB	-17.1	-14.2	-2.8	3.4	10.7	18.7	20.0	17.5	13.4	1.2	-7.8	-17.3
	2%	DB	-21.6	-17.5	-4.1	5.1	15.4	24.7	27.0	22.5	14.0	0.5	-10.0	-19.5
		MCWB	-22.0	-18.0	-5.4	1.3	8.4	16.3	19.3	16.4	10.1	-0.3	-10.5	-19.9
	5%	DB	-24.6	-20.7	-6.8	3.1	12.6	22.4	24.7	20.4	11.4	-0.7	-13.1	-22.5
		MCWB	-24.9	-21.1	-7.9	0.3	7.2	15.0	18.5	15.2	8.5	-1.4	-13.6	-22.7
	10%	DB	-26.7	-24.2	-9.5	1.4	9.8	20.1	22.4	18.3	9.5	-2.0	-17.6	-25.4
		MCWB	-26.9	-24.5	-10.4	-1.0	5.3	14.1	17.4	14.2	7.0	-2.7	-18.0	-25.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-17.1	-14.2	-2.7	3.5	11.0	20.0	21.7	19.2	13.6	1.5	-7.6	-17.3
		MCDB	-16.7	-13.5	-0.9	8.0	18.7	24.9	27.1	23.3	17.6	1.8	-6.9	-17.0
	2%	WB	-21.9	-18.0	-5.2	2.0	9.0	17.6	20.3	17.2	10.9	-0.1	-10.5	-19.8
		MCDB	-21.5	-17.5	-4.1	4.5	14.6	22.9	25.2	20.6	13.3	0.3	-10.0	-19.5
	5%	WB	-24.9	-21.1	-7.9	0.6	7.5	16.1	19.1	15.7	8.9	-1.4	-13.5	-22.7
		MCDB	-24.7	-20.8	-6.8	2.9	12.1	20.6	23.5	19.3	11.0	-0.7	-13.0	-22.5
	10%	WB	-26.9	-24.4	-10.4	-1.1	5.7	14.6	17.9	14.5	7.2	-2.7	-17.9	-25.7
		MCDB	-26.7	-24.2	-9.5	1.4	9.4	19.3	21.8	17.7	9.1	-2.0	-17.6	-25.4
Mean Daily Temperature Range		MDBR	6.0	7.2	10.4	11.0	8.5	9.0	8.7	8.6	6.9	5.7	6.3	5.7
	5% DB	MCDBR	7.5	9.2	8.9	10.2	11.7	12.7	12.1	12.7	11.2	4.3	7.5	8.0
		MCWBR	7.4	8.8	8.1	7.6	6.8	6.5	6.3	7.3	7.4	4.0	7.4	7.8
	5% WB	MCDBR	7.5	9.2	8.8	9.0	11.1	10.9	10.5	11.1	10.2	4.2	7.5	8.1
		MCWBR	7.3	8.8	8.0	7.0	6.7	6.8	6.1	6.8	7.3	4.1	7.4	7.8
Clear-Sky Solar Irradiance	taub	0.185	0.225	0.284	0.319	0.353	0.363	0.444	0.422	0.308	0.238	0.181	N/A	
	taud	1.873	2.022	2.060	2.081	2.205	2.418	2.155	2.171	2.472	2.121	1.888	N/A	
	Ebn at Noon	57	547	723	811	827	835	731	696	733	519	46	0	
	Edh at Noon	6	46	86	115	115	97	123	108	60	40	5	0	
All-Sky Solar Radiation	RadAvg	0.03	0.55	2.23	4.49	5.75	5.75	5.17	3.43	1.88	0.59	0.09	0.00	
	RadStd	0.00	0.04	0.14	0.20	0.47	0.60	0.42	0.41	0.18	0.09	0.01	0.00	